

Robot Framework

Assignment 2: Variables and Data-Driven Automation

1. Problem Statement: -

Implement a test case that uses variables to store data like usernames and passwords. Create a data-driven test that reads test data from an external file and performs the same set of actions on multiple data sets.

2. Objective: -

Implement data-driven testing in Robot Framework using Variables and Test Templates.

3. Tools, Software, and Concepts Used: -

Robot Framework, SeleniumLibrary, DataDriver Library (Optional/External CSV reading).

4. Source Code / Implementation: -

```
*** Settings ***
Library                SeleniumLibrary
Test Setup             Open Browser To Login Page
Test Teardown          Close Browser
Test Template          Login With Invalid Credentials Should Fail

*** Variables ***
${LOGIN_URL}           https://www.saucedemo.com/
${BROWSER}             Chrome

*** Test Cases ***
                        USERNAME      PASSWORD
Invalid Username      invalid_user   secret_sauce
Invalid Password      standard_user  wrong_pass
Empty Credentials     ${EMPTY}      ${EMPTY}

*** Keywords ***
Open Browser To Login Page
    Open Browser      ${LOGIN_URL}    ${BROWSER}
    Maximize Browser Window
    Sleep             2s
Login With Invalid Credentials Should Fail
    [Arguments]       ${username}    ${password}
    Input Text        id:user-name   ${username}
    Input Text        id:password    ${password}
    Sleep             3s
    Click Element     id:login-button
    Page Should Contain Element      css:.error-message-container
```

5. Output & Screenshots: -

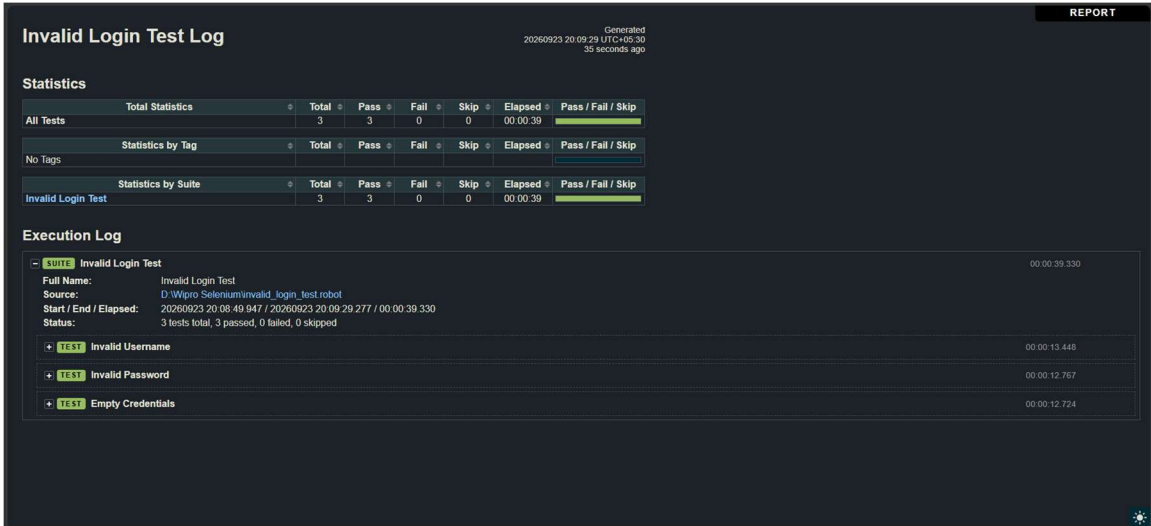


Fig.1: Invalid Login Test Log

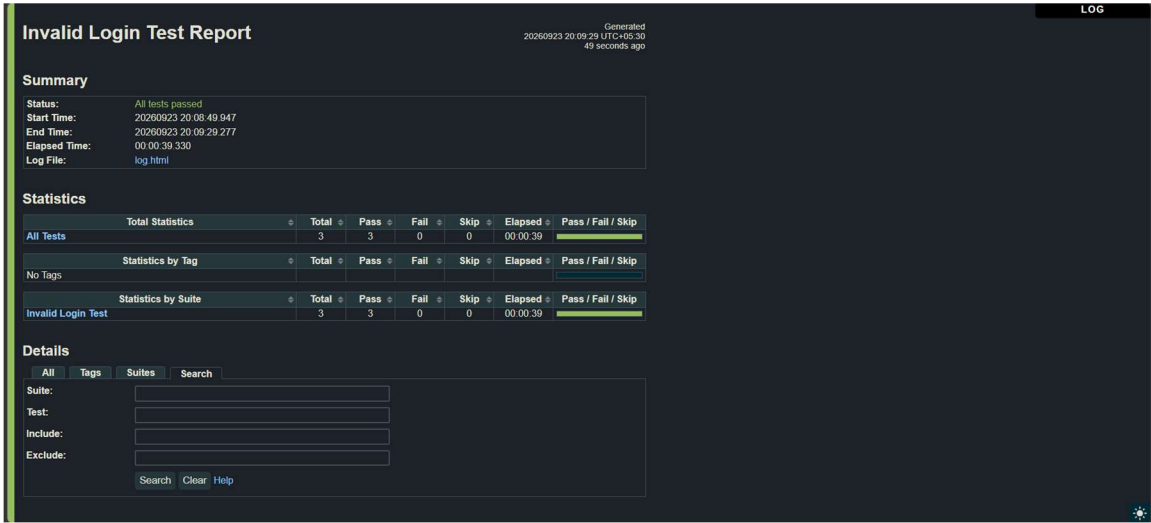


Fig.2: Invalid Login Test Report

```

PS D:\Wipro Selenium> robot invalid_login_test.robot
=====
Invalid Login Test
=====
Invalid Username | PASS |
-----
Invalid Password | PASS |
-----
Empty Credentials | PASS |
-----
Invalid Login Test | PASS |
3 tests, 3 passed, 0 failed
=====
Output: D:\Wipro Selenium\output.xml
Log: D:\Wipro Selenium\log.html
Report: D:\Wipro Selenium\report.html

```

Fig.3: Terminal output

6. Result, Observation, and Conclusion: -

- Result: The test template executed multiple permutations of the login sequence using different inline variables.
- Observation: Data-driven tests in Robot Framework drastically reduce script size by reusing a single block of keyword logic for multiple data rows.
- Conclusion: Parameterizing variables and templates is critical for efficient regression testing.